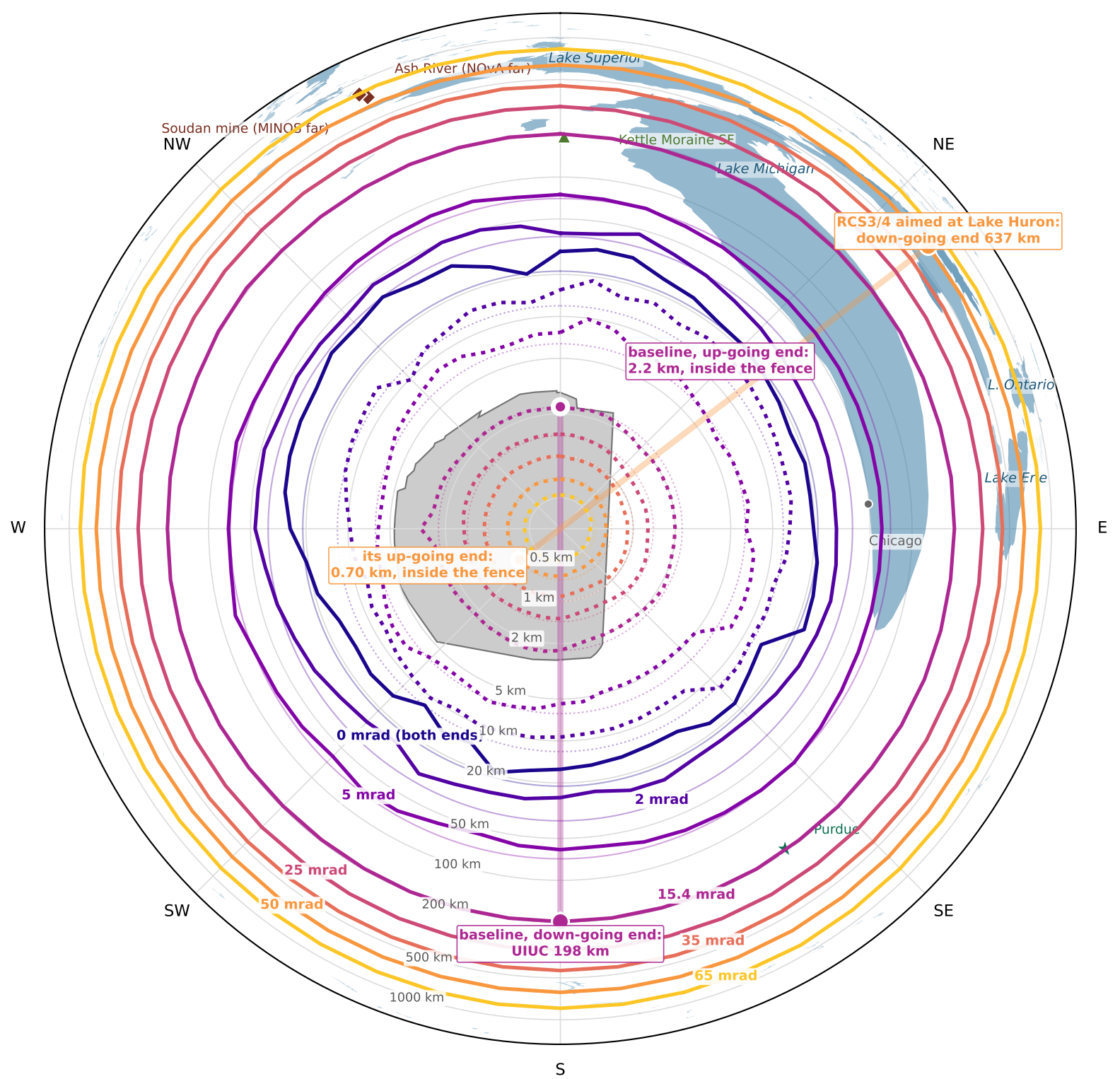


Where the two ends of one straight surface, tilt by tilt
35 m-deep straight; one colour per tilt – dotted: the end that goes up, solid: the end that goes down; at 0 mrad they are the same curve
N



- up-going end surfaces here (near emergence)
- same, smooth sphere
- lakes (Natural Earth 10m)
- down-going end surfaces here (far exit)
- Fermilab site
- one straight: its two ends are opposite each other

Bold curves: GEBCO 2020 terrain along 48 bearings (0.25–1000 km), exits over the Great Lakes taken at the water surface. Faint circles: smooth sphere,
 $s_{near} = R_E(-\theta + \sqrt{\theta^2 + 2d_0/R_E}) \approx d_0/\theta$ and $s_{far} = R_E(\theta + \sqrt{\theta^2 + 2d_0/R_E}) \approx 2R_E\theta$. Log range; rings in km. The fence is 0.9–4.9 km out, so the dotted curves show which tilts emerge on site.